Fundamentals of Algebraic Expressions

Comprehensive Solutions to Exercise 13C, Mental Maths, & Competency Questions

Generated by Gemini Notebook — Grounded in Chapter 13 Text Materials

1 Exercise 13C — Multiple Choice Questions

Choose the correct option in each of the following:

Q1. Which of the following is a literal?

(a) 0.5

(b) 0

(c) s

(d) $\frac{3}{7}$

Solution: By definition, a *literal* (or variable) is a symbol or letter used to represent an unknown value that can vary across different problems. On the other hand, quantities denoted by figures with a fixed value (such as 0.5, 0, and $\frac{3}{7}$) are *constants*. Therefore, s is the literal. □

Correct Option / Answer: (c) s

Q2. The coefficient of x in $-6xyz^2$ is:

(a) -6

(b) $-6x$

(c) yz^2

(d) $-6yz^2$

Solution: The coefficient of any factor in a term is the product of all the remaining factors of that term. To find the coefficient of x in $-6xyz^2$, we divide the term by x:

$$\text{Coefficient of } x = \frac{-6xyz^2}{x} = -6yz^2$$

This represents the literal and numerical factors that multiply x. □

Correct Option / Answer: (d) $-6yz^2$

Q3. Which of the following is a pair of like terms?

(a) ab, xy

(b) $2xz^2, -7x^2z$

(c) $4a^2b^2, -\frac{1}{3}a^2bc$

(d) $0.9xy, -\frac{2}{11}xy$

Solution: Like terms are terms that have the exact same literal factors (or algebraic part) with identical exponents on corresponding variables.

- (a) ab and xy have different literal parts.
- (b) $2xz^2$ has variables x^1z^2, while $-7x^2z$ has variables x^2z^1. The exponents do not match.
- (c) $4a^2b^2$ has literal part a^2b^2, while $-\frac{1}{3}a^2bc$ has a^2bc.
- (d) $0.9xy$ and $-\frac{2}{11}xy$ both have the identical literal factor xy.

Thus, (d) is the only pair of like terms. □

Correct Option / Answer: **(d)** $0.9xy, -\frac{2}{11}xy$

Q4. The degree of the polynomial $ma^2b^3 + 6ab^2 - m$ is:

(a) 6

(b) 5

(c) 3

(d) 2

Solution: The degree of a term in a polynomial is the sum of the powers of all the variables involved in that term. The degree of the polynomial is the highest degree among all its terms. Let us analyze the terms of $ma^2b^3 + 6ab^2 - m$:

- If a, b, and m are all treated as variables:
  - Term 1 (ma^2b^3): The sum of the exponents of m^1, a^2, and b^3 is $1 + 2 + 3 = 6$.
  - Term 2 ($6ab^2$): The sum of exponents of a^1 and b^2 is $1 + 2 = 3$.
  - Term 3 ($-m^1$): The exponent of m is 1.

  The highest term degree is 6.

- If m is treated as a constant parameter (with only a and b as variables):
  - Term 1 (ma^2b^3): The degree in a and b is $2 + 3 = 5$.
  - Term 2 ($6ab^2$): The degree in a and b is $1 + 2 = 3$.

  In this case, the degree would be 5.

In standard algebraic work, unless explicitly designated otherwise, all literal letters (such as a, b, m) are counted as variables, making the degree **6**. We present both interpretations for absolute mathematical rigor. □

Correct Option / Answer: (a) 6 (or (b) 5 if m is a constant coefficient)

Q5. The expression $-3h^4 + 7(h - 1)$ is a:

(a) monomial

(b) binomial

(c) trinomial

(d) none of these

Solution: First, let us simplify the algebraic expression by expanding the parentheses:

$$-3h^4 + 7(h - 1) = -3h^4 + 7h - 7$$

This simplified expression contains three distinct, unlike terms: $-3h^4$, $7h$, and -7. An algebraic expression containing exactly three terms is called a **trinomial**. □

Correct Option / Answer: (c) trinomial

Q6. Which of the following is a binomial?

(a) $a^2 - 6b^2 + 1000$

(b) $-7x^2y + xy^2 + 3x^2y$

(c) $ma^2b - (m+n)a^2b^2 + nab^2$

(d) $st^2 - st + 1$

Solution: A binomial is an algebraic expression containing exactly two unlike terms. Let us evaluate each option:

- (a) $a^2 - 6b^2 + 1000$ has 3 unlike terms (trinomial).
- (b) $-7x^2y + xy^2 + 3x^2y$: Here, $-7x^2y$ and $3x^2y$ are like terms and must be combined:

  $$(-7x^2y + 3x^2y) + xy^2 = -4x^2y + xy^2$$

  This simplified expression has exactly 2 unlike terms, so it is a binomial.
- (c) $ma^2b - (m+n)a^2b^2 + nab^2$ has 3 unlike terms (trinomial).
- (d) $st^2 - st + 1$ has 3 unlike terms (trinomial).

Thus, option (b) simplifies to a binomial. □

Correct Option / Answer: (b) $-7x^2y + xy^2 + 3x^2y$

Q7. Which of the following is a polynomial?

(a) $x - \frac{1}{x}$

(b) $x + \sqrt{x}$

(c) $\frac{x}{\sqrt{2}} - \frac{1}{\sqrt{2}}$

(d) $\frac{1}{x^2} + x^2$

Solution: By definition, a polynomial is an algebraic expression in which all variables are raised only to non-negative integer powers (i.e., $0, 1, 2, 3, \ldots$).

- (a) $x - \frac{1}{x} = x - x^{-1}$: The exponent is -1 (negative integer), so it is not a polynomial.
- (b) $x + \sqrt{x} = x + x^{1/2}$: The exponent is $\frac{1}{2}$ (not an integer), so it is not a polynomial.
- (c) $\frac{x}{\sqrt{2}} - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}}x^1 - \frac{1}{\sqrt{2}}$: The exponent of the variable x is 1, which is a non-negative integer. Note that coefficients can be any real numbers (such as $\frac{1}{\sqrt{2}}$). Thus, this is a valid polynomial.
- (d) $\frac{1}{x^2} + x^2 = x^{-2} + x^2$: The exponent -2 is negative, so it is not a polynomial.

□

Correct Option / Answer: (c) $\frac{x}{\sqrt{2}} - \frac{1}{\sqrt{2}}$

Q8. Which of the following has the highest degree?

(a) $x^5 + 2x^5$

(b) 2^8

(c) $\frac{3a^3b^2c}{4}$

(d) $\frac{7a^6b^5}{-5a^3b^4}$

Solution: Let us simplify and calculate the degree of each expression:

- (a) $x^5 + 2x^5 = 3x^5$: The exponent of the variable x is 5, so the degree is 5.
- (b) $2^8 = 256$: This is a constant term (no variable). The degree of a non-zero constant is 0.
- (c) $\frac{3}{4}a^3b^2c^1$: The degree is the sum of exponents of the variables: $3 + 2 + 1 = 6$.
- (d) $\frac{7a^6b^5}{-5a^3b^4} = -1.4a^{6-3}b^{5-4} = -1.4a^3b^1$: The degree of this simplified monomial is $3 + 1 = 4$.

Comparing the degrees: 5, 0, 6, and 4, we find that option (c) has the highest degree of 6. □

Correct Option / Answer: (c) $\frac{3a^3b^2c}{4}$

Q9. The value of the expression $2m^2 + 3n^2m + \frac{7}{2}n^2 + 8$, when $m = -2$ and $n = 2$ is:

(a) 1

(b) -4

(c) 6

(d) -14

Solution: Substitute $m = -2$ and $n = 2$ directly into the expression:

$$\text{Value} = 2(-2)^2 + 3(2)^2(-2) + \frac{7}{2}(2)^2 + 8$$

Evaluate term by term:

1. $2(-2)^2 = 2 \times 4 = 8$
2. $3(2)^2(-2) = 3 \times 4 \times (-2) = -24$
3. $\frac{7}{2}(2)^2 = \frac{7}{2} \times 4 = 14$
4. Constant term $= 8$

Now sum these results:

$$\text{Value} = 8 - 24 + 14 + 8 = 6$$

□

Correct Option / Answer: (c) 6

Q10. The sum of the polynomials $bn - 8am - 8cp$, $5cp - am + 2bn$ and $6am + 5bn - 3cp$ is equal to:

(a) $-2am + 2bn - 4cp$

(b) $-2am + 8bn - 2cp$

(c) $-3am + 2bn - 4cp$

(d) $-3am + 8bn - 6cp$

Solution: To sum the polynomials, we write them down and group the like terms together (column-wise or group-wise):

$$\text{Sum} = (bn - 8am - 8cp) + (5cp - am + 2bn) + (6am + 5bn - 3cp)$$

Group by each literal part:

- Terms in am: $-8am - am + 6am = (-8 - 1 + 6)am = -3am$
- Terms in bn: $bn + 2bn + 5bn = (1 + 2 + 5)bn = 8bn$
- Terms in cp: $-8cp + 5cp - 3cp = (-8 + 5 - 3)cp = -6cp$

Combining these groups gives:

$$\text{Sum} = -3am + 8bn - 6cp$$

□

Correct Option / Answer: (d) $-3am + 8bn - 6cp$

2 Mental Maths

1. Fill in the blanks:

(i) The highest power of the **variable** in a polynomial is called its degree.

(ii) The degree of the polynomial 7^5 is **0**.
Solution: Since $7^5 = 16807$ is a constant numerical value (which has no variable component), its degree is 0. Exponents on constant numbers do not determine a polynomial's degree. □

(iii) Several parts of an algebraic expression separated by $+$ or $-$ signs are called the **terms** of the expression.

(iv) Any number is a polynomial of degree **0**. (This refers to non-zero constant numbers, which are constant polynomials).

(v) Terms having the same **literal coefficients** (or algebraic factors) are called like terms.

(vi) The algebraic expression for the statement 'the number of times b is contained in x' is $\boldsymbol{\frac{x}{b}}$ (or $\boldsymbol{x \div b}$).

(vii) $1 + x + y + xy$ is a polynomial having **4** terms and degree **2**.
Solution: The terms are 1, x, y, and xy. The term with the highest degree is xy, which has degree $1 + 1 = 2$. □

(viii) The length of a side of a square having perimeter $8x^2 - 2y + 16xy$ is $\underline{\boldsymbol{2x^2 - \frac{1}{2}y + 4xy}}$.
Solution: Since Perimeter $= 4 \times$ side, we have:

$$\text{side} = \frac{\text{Perimeter}}{4} = \frac{8x^2 - 2y + 16xy}{4} = 2x^2 - \frac{1}{2}y + 4xy$$

□

(ix) In a polynomial, the exponents of the variables are always **<u>non-negative integers</u>** (or whole numbers).

(x) If the length of a rectangle having perimeter $(6m^2 - 2mn + 2m^2n - 4n^2)$ units, is $(3m^2 + m^2n - 2mn)$ units, then its breadth is equal to $\underline{\boldsymbol{mn - 2n^2}}$ units.
Solution: We know that $P = 2(L + B) \implies L + B = \frac{P}{2}$.

$$\frac{P}{2} = \frac{6m^2 - 2mn + 2m^2n - 4n^2}{2} = 3m^2 - mn + m^2n - 2n^2$$

Subtracting the length $L = 3m^2 + m^2n - 2mn$ to find the breadth B:

$$B = \left(3m^2 - mn + m^2n - 2n^2\right) - \left(3m^2 + m^2n - 2mn\right)$$

$$B = 3m^2 - 3m^2 + m^2n - m^2n - mn - (-2mn) - 2n^2 = mn - 2n^2 \text{ units}$$

□

2. Write true (T) or false (F):

(i) A literal can take on various numerical values. [**True (T)**]
Reason: By definition, a literal/variable is a symbol that represents varying numerical values.

(ii) $2a^3b - a^2b - 3a^2b^2 + 7ba^2 - ba^3$ is a trinomial. [**True (T)**]
Solution: We must simplify the expression by combining like terms:

- Terms with a^3b: $2a^3b - ba^3 = a^3b$
- Terms with a^2b: $-a^2b + 7ba^2 = 6a^2b$
- Term with a^2b^2: $-3a^2b^2$

The simplified expression is $a^3b + 6a^2b - 3a^2b^2$, which consists of exactly three unlike terms. Thus, it is a trinomial. □

(iii) $3mn$ is a factor of $-9mn^2$. [**True (T)**]
Reason: Since $-9mn^2 = 3mn \times (-3n)$, $3mn$ divides the term completely and is thus a factor.

(iv) If we add a monomial and a trinomial, the answer can be a monomial. [**False (F)**]
Solution: A trinomial contains exactly 3 unlike terms. Adding a monomial can at most combine with one of those terms (if they are like terms), reducing the expression to $3 - 1 + 1 = 3$ terms, or it can be unlike all of them, resulting in 4 terms. It can never cancel out two terms of the trinomial to leave a single term (monomial). □

(v) The coefficient of a^2b in $-9a^2b^2c$ is $-9bc$. [**True (T)**]
Reason: Factoring out a^2b from $-9a^2b^2c$ yields: $-9a^2b^2c = (a^2b) \cdot (-9bc)$.

(vi) The degree of the monomial 3^3 is 3. [**False (F)**]
Reason: $3^3 = 27$ is a non-zero constant. The degree of any non-zero constant polynomial is 0.

3 Case Study Based Questions

Case Study I

Saket wrote four different algebraic expressions in his notebook:

$$f(x) = 2x^3 - \frac{1}{3}x, \quad g(x) = x - \frac{1}{x} + 2, \quad h(x) = x^2 - 2\sqrt{x}, \quad k(y) = 2y - 1$$

1. Which of these expressions is/are binomials?

(a) $f(x)$, $g(x)$, $h(x)$ only

(b) $f(x)$, $g(x)$, $k(y)$ only

(c) $f(x)$, $h(x)$, $k(y)$ only

(d) $f(x)$, $h(x)$ only

Solution: A binomial is an algebraic expression containing exactly 2 terms.

- $f(x) = 2x^3 - \frac{1}{3}x$ has 2 terms.
- $g(x) = x - \frac{1}{x} + 2$ has 3 terms (trinomial).
- $h(x) = x^2 - 2\sqrt{x}$ has 2 terms.
- $k(y) = 2y - 1$ has 2 terms.

Thus, $f(x)$, $h(x)$, and $k(y)$ are binomials. □

Correct Option / Answer: (c) $f(x)$, $h(x)$, $k(y)$ only

2. Which of these expressions is/are not polynomials?

(a) $f(x)$, $g(x)$, $h(x)$ only

(b) $f(x)$, $h(x)$ only

(c) $g(x)$, $h(x)$ only

(d) $h(x)$ only

Solution: Let us check the exponents of the variables:

- In $g(x) = x - x^{-1} + 2$, the term x^{-1} has a negative integer exponent.
- In $h(x) = x^2 - 2x^{1/2}$, the term $2\sqrt{x} = 2x^{1/2}$ has a fractional exponent.

Since polynomials must only have non-negative integer exponents, $g(x)$ and $h(x)$ are not polynomials. Note that $f(x)$ (exponents 3, 1) and $k(y)$ (exponent 1) are polynomials. □

Correct Option / Answer: (c) $g(x)$, $h(x)$ only

3. Which of these expressions is/are polynomials in two variables?

(a) $f(x)$ only

(b) $k(y)$ only

(c) $f(x)$, $k(y)$ only

(d) None of these

Solution: All of Saket's expressions involve only a single variable: $f(x)$, $g(x)$, and $h(x)$ contain only the variable x, and $k(y)$ contains only the variable y. Therefore, none of them are in two variables. □

Correct Option / Answer: (d) None of these

4. The degree of the polynomial $f(x)$ is:

(a) 0

(b) 1

(c) 2

(d) 3

Solution: For $f(x) = 2x^3 - \frac{1}{3}x$, the highest exponent of the variable x is 3 (in the first term $2x^3$). Hence, the degree is 3. □

Correct Option / Answer: (d) 3

Case Study II

Vasu has a rectangular farmland A. The length of this farmland is $x^2y - 2xy + 3x^2$ and its perimeter is $2x^2y + 6x^2 - 2xy - 4y^2$. Today he purchased the adjacent farmland B and combined the two farmlands into one. The length of his farmland now increased by $x^2 + xy$, while the breadth remained the same.

1. Find the breadth of farmland A:

(a) $x^2y + 3x^2 - 4y^2$

(b) $xy - 2y^2$

(c) $12x^2 - 4xy$

(d) $6x^2 + xy - 2y^2$

Solution: The perimeter of a rectangle is $P = 2(L + B) \implies L + B = \frac{P}{2}$. Given $P_A = 2x^2y + 6x^2 - 2xy - 4y^2$, we calculate half of the perimeter:

$$L_A + B_A = \frac{2x^2y + 6x^2 - 2xy - 4y^2}{2} = x^2y + 3x^2 - xy - 2y^2$$

To find the breadth B_A, we subtract the length $L_A = x^2y - 2xy + 3x^2$:

$$B_A = (L_A + B_A) - L_A = (x^2y + 3x^2 - xy - 2y^2) - (x^2y - 2xy + 3x^2)$$

Aligning like terms for column subtraction:

$$\begin{array}{rcccc} (L_A + B_A): & x^2y & +3x^2 & -xy & -2y^2 \\ -L_A: & -(x^2y & +3x^2 & -2xy) & \\ \hline B_A: & 0 & +0 & +xy & -2y^2 \end{array}$$

Thus, the breadth of farmland A is $xy - 2y^2$. □

Correct Option / Answer: (b) $xy - 2y^2$

2. Find the length of the combined farmland owned by Vasu:

(a) $x^2 + 2xy - 2y^2$

(b) $4x^2 - 3xy + x^2y$

(c) $4x^2 - xy + x^2y$

(d) $2x^2 - 3xy - 2y^2$

Solution: The combined length is the original length L_A plus the increase $x^2 + xy$:

$$L_{\text{comb}} = L_A + (x^2 + xy) = (x^2y - 2xy + 3x^2) + (x^2 + xy)$$

Combine like terms:

$$L_{\text{comb}} = x^2y + (-2xy + xy) + (3x^2 + x^2) = x^2y - xy + 4x^2$$

Rearranging terms: $4x^2 - xy + x^2y$. □

Correct Option / Answer: (c) $4x^2 - xy + x^2y$

3. The perimeter of Vasu's combined farmland is:

(a) $8x^2 + 2x^2y - 4y^2$

(b) $4y^2 + 4xy + 2x^2y + 8x^2$

(c) $4x^2 - 2y^2 + 4y^2$

(d) $4x^2 + 4xy - 2y^2 - 8x^2$

Solution: The combined perimeter is $P_{\text{comb}} = 2(L_{\text{comb}} + B_{\text{comb}})$. Since the breadth remains unchanged, $B_{\text{comb}} = B_A = xy - 2y^2$. First, calculate $L_{\text{comb}} + B_{\text{comb}}$:

$$L_{\text{comb}} + B_{\text{comb}} = (4x^2 - xy + x^2y) + (xy - 2y^2) = 4x^2 + x^2y - 2y^2$$

Now multiply by 2:

$$P_{\text{comb}} = 2(4x^2 + x^2y - 2y^2) = 8x^2 + 2x^2y - 4y^2$$

□

Correct Option / Answer: **(a)** $8x^2 + 2x^2y - 4y^2$

4. The change in the perimeter after combining farmlands A and B is:

(a) $4x^2 + 2xy + 4x^2y - y^2$

(b) $2x^2 - 2xy + y^2$

(c) $2x^2 + 2xy - y^2$

(d) $2x^2 + 2xy$

Solution: The change in perimeter is the new perimeter minus the original perimeter:

$$\text{Change} = P_{\text{comb}} - P_A$$

$$\text{Change} = (8x^2 + 2x^2y - 4y^2) - (2x^2y + 6x^2 - 2xy - 4y^2)$$

Distribute the negative sign and group like terms:

$$\text{Change} = 8x^2 - 6x^2 + 2x^2y - 2x^2y - 4y^2 - (-4y^2) - (-2xy)$$

$$\text{Change} = 2x^2 + 2xy$$

□

Correct Option / Answer: (d) $2x^2 + 2xy$

4 Assertion-Reason & Competency Focused Questions

1. Assertion-Reason Questions

Q1. Assertion (A): The expression $x + y - 2x$ is a trinomial.
Reason (R): An algebraic expression containing three terms is called a trinomial.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(c) (A) is true but (R) is false.
(d) (A) is false but (R) is true.

Solution:

- Let us simplify the expression in (A): $x + y - 2x = y - x$. This simplified expression has exactly 2 terms, so it is a binomial, not a trinomial. Thus, Assertion (A) is **False**.
- Reason (R) states the standard definition of a trinomial, which is correct. Thus, Reason (R) is **True**.

□

Correct Option / Answer: (d) (A) is false but (R) is true.

Q1. Assertion (A): If we subtract $15x^2 - 9x + 1$ from 1, then we get $-15x^2 + 9x$.
Reason (R): The degree of $15x^2 - 9x + 1$ is 3 as it has three terms and degree of $-15x^2 + 9x$ is 2 as it has two terms.

(a) Both (A) and (R) are true and (R) is the correct explanation of (A).

(b) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(c) (A) is true but (R) is false.

(d) (A) is false but (R) is true.

Solution:

- Let us perform the subtraction in (A):

$$1 - (15x^2 - 9x + 1) = 1 - 15x^2 + 9x - 1 = -15x^2 + 9x$$

  This matches the statement, so Assertion (A) is **True**.

- In (R), the expression $15x^2 - 9x + 1$ has the highest power of x as 2, so its degree is 2, not 3. (The number of terms has no bearing on its degree). Hence, Reason (R) is **False**.

□

Correct Option / Answer: (c) (A) is true but (R) is false.

2. Competency Focused Questions

Q1. Consider the expression: "5 subtracted from the sum of 5 times u squared and 5 times the product of u and -1". Which of these statements is correct?

(a) All terms of the expression are: $5u^2$, $-5u$ and -5.

(b) All terms of the expression are: $5u^2$ and $-5u$.

(c) All factors of the term $-5u$ are: 5 and u.

(d) All factors of the term $5u^2$ are: 5 and u.

Solution: Let us translate the word statement into an algebraic expression step-by-step:

- "5 times u squared": $5u^2$
- "5 times the product of u and -1": $5(u \times -1) = -5u$
- "sum of [these]": $5u^2 + (-5u) = 5u^2 - 5u$
- "5 subtracted from [this sum]": $(5u^2 - 5u) - 5 = 5u^2 - 5u - 5$

The parts of an algebraic expression separated by $+$ or $-$ signs are its terms. The terms of this expression are $5u^2$, $-5u$, and -5. This matches option (a) exactly. □

Correct Option / Answer: (a) All terms of the expression are: $5u^2$, $-5u$ and -5.

Q1. Study the following statements and select the correct option.
P: The number of unlike terms in $4pq^2 - 2q^2r + q^2p + 3pqr + q^2r - 8$ is 4.
Q: An expression having three or more terms is called a trinomial.

(a) Both P and Q are true

(b) Both P and Q are false

(c) P is true but Q is false

(d) P is false but Q is true

Solution:

- Let us simplify the expression in Statement P:

$$4pq^2 - 2q^2r + q^2p + 3pqr + q^2r - 8$$

  Combine like terms:

  – $4pq^2$ and q^2p (since $q^2p = pq^2$): $4pq^2 + q^2p = 5pq^2$
  – $-2q^2r$ and q^2r: $-2q^2r + q^2r = -q^2r$
  – Unlike terms remaining: $3pqr$ and -8.

  The fully simplified expression is $5pq^2 - q^2r + 3pqr - 8$. This has exactly 4 unlike terms, so Statement P is **True**.

- Trinomial is defined specifically as an expression containing exactly three terms, not "three or more". (Expressions with more terms are generally called multinomials). Thus, Statement Q is **False**.

□

Correct Option / Answer: (c) P is true but Q is false

Q1. The cost of entry tickets to an amusement park are different for men and children. A man's ticket costs Rs 800 and a child's ticket costs Rs 500. On Monday, the number of men who visited the amusement park was the square of the number of children who visited. How much money was collected by selling entry tickets on Monday?

(a) x^2

(b) $800x^2 + 500x$

(c) $800x + 500x^2$

(d) $1300(x + x^2)$

Solution: Let x represent the number of children who visited the park. Then, the number of men who visited is x^2.

- Revenue from child tickets = Cost per child × Number of children = $500x$
- Revenue from man tickets = Cost per man × Number of men = $800x^2$

Total revenue collected is the sum of both: $800x^2 + 500x$. □

Correct Option / Answer: **(b)** $800x^2 + 500x$

Q1. Sonika and Komal are collecting some marbles for their project. Sonika collected 15 red marbles and $6x$ blue marbles, whereas Komal collected $5y$ red marbles and $8x$ blue marbles. After some time Sonika loses $3y$ marbles and Komal collected 8 more marbles. How many total number of marbles they both have now, if $x = 2$ and $y = 3$?

(a) 54
(b) 65
(c) 57
(d) 62

Solution: Let us write down the total number of marbles step-by-step:

- Sonika's marbles $= 15 + 6x$
- Komal's marbles $= 5y + 8x$
- Initial Combined Marbles $= (15 + 6x) + (5y + 8x) = 15 + 14x + 5y$
- Sonika loses $3y$ marbles: New Total $= (15 + 14x + 5y) - 3y = 15 + 14x + 2y$
- Komal collects 8 more marbles: Final Total $= (15 + 14x + 2y) + 8 = 23 + 14x + 2y$

Now substitute $x = 2$ and $y = 3$:

$$\text{Final Total} = 23 + 14(2) + 2(3) = 23 + 28 + 6 = 57 \text{ marbles}$$

□

Correct Option / Answer: (c) 57

Q1. What is the sum of the numerical coefficients of the variables m and n in the expression given by the statement? "Circumference of the circle of radius m subtracted from the circumference of the circle of radius $\frac{n}{2}$".

(a) $-\pi$
(b) -1
(c) 3
(d) π

Solution: Let us construct the algebraic expression:

- Circumference of circle of radius m: $C_m = 2\pi m$
- Circumference of circle of radius $\frac{n}{2}$: $C_{n/2} = 2\pi\left(\frac{n}{2}\right) = \pi n$
- "Circumference of circle of radius m subtracted from the other":

$$\text{Expression} = \pi n - 2\pi m = -2\pi m + \pi n$$

The variables are m and n:

- The numerical coefficient of m is -2π.
- The numerical coefficient of n is π.

Sum of the numerical coefficients $= -2\pi + \pi = -\pi$. □

Correct Option / Answer: (a) $-\pi$

Q1. Which of the following statement is true about the expression given below?

$$-14m^2l - 17m^2 + 4n^2 - 15lm^2 - (-6m^2 - 10n^2 + 18l^2m^2) - 12m^2l^2$$

(a) The simplified form is a trinomial $-59m^2l - 11m^2 + 14n^2$.

(b) The simplified form is a trinomial $-35m^2l - 23m^2 + 6n^2$.

(c) The simplified form is a polynomial $-29m^2l - 23m^2 - 6n^2 + 6l^2m^2$.

(d) The simplified form is a polynomial $-29m^2l - 11m^2 + 14n^2 - 30l^2m^2$.

Solution: Let us remove the parentheses and group like terms of the expression:

$$-14m^2l - 17m^2 + 4n^2 - 15lm^2 + 6m^2 + 10n^2 - 18l^2m^2 - 12m^2l^2$$

Now, identify and combine the like terms:

- Terms with m^2l (note $lm^2 = m^2l$): $-14m^2l - 15lm^2 = (-14 - 15)m^2l = -29m^2l$
- Terms with m^2: $-17m^2 + 6m^2 = (-17 + 6)m^2 = -11m^2$
- Terms with n^2: $4n^2 + 10n^2 = (4 + 10)n^2 = 14n^2$
- Terms with l^2m^2 (note $m^2l^2 = l^2m^2$): $-18l^2m^2 - 12m^2l^2 = (-18 - 12)l^2m^2 = -30l^2m^2$

Combining these simplified terms, we get:

$$\text{Simplified Expression} = -29m^2l - 11m^2 + 14n^2 - 30l^2m^2$$

This is a polynomial containing four unlike terms. This matches option (d) exactly. □

Correct Option / Answer: (d) The simplified form is a polynomial $-29m^2l - 11m^2 + 14n^2 - 30l^2m^2$.